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%Task 1
s = linspace(0, 2, 201);
h = linspace(0, 2,201);
L = []; % Unbounded
K = []; % Bounded
N = 1000;
x = 1 : 1: 1000;
v = zeros(1, max(N));
% s = .22;
% h = .75;
a = -0.2j;
b = -0.4;
d = 0.9;
v(1) = 0;
v(2) = 0;
v(3) = -0.2 - 0.1 * j;
flag = 0; % Flag ensures that values aren't added to both the K and L array
for m= 1:size(s,2) % iterates every combination of h and s
    for y = 1:size(h,2)
        flag = 0;
        c = s(m) + j*h(y);
        for n = 2:N
            v(n+2) = v(n+1)^2 + d*v(n)^3 + a*v(n-1) + b*c;
            if(abs(v(n+2)) > 10)
                L(end +1) = c;
                flag = 1;
                break;
            end
        end
       if(flag == 0)
           K(end +1) = c;
       end
    end
end

figure(1);
clf;
plot(real(K), imag(K), 'b.', 'DisplayName', 'Bounded (safe)');
hold on;
plot(real(L), imag(L), 'r.', 'DisplayName', 'Unbounded (unsafe)');
axis square;
xlabel('SiO2 concentration (s)');
```

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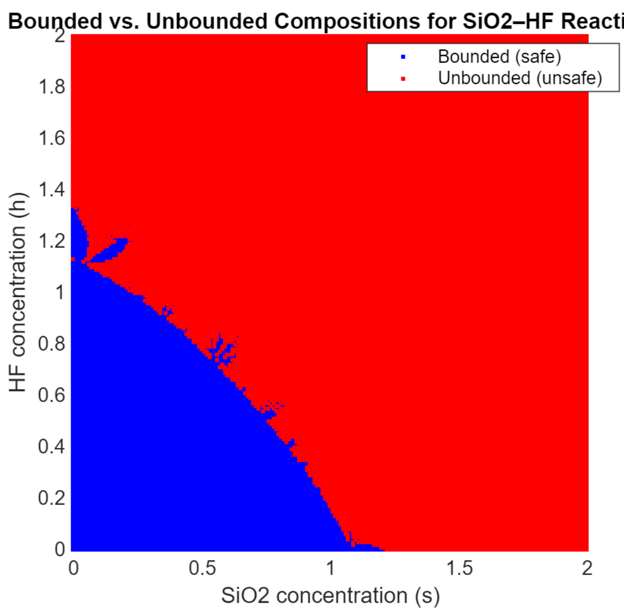
ylabel('HF concentration (h)');
title('Bounded vs. Unbounded Compositions for SiO2-HF Reaction');
legend('Location','best');
grid on;
```

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% Task 2:
% The numbers provided by the hot shot scientist are wildy innacurate.
% After simulating the equation   v(n+2) = v(n+1)^2 + d*v(n)^3 + a*v(n-1) +
b*c;
%  where
a = -0.2;
b = -0.4j;
d = 0.9;
s = .22;
h = .75;
c = s + j*h;
% with the max value of n being 1000 we can see that the value of
% v by the end of the 77th n index surpasses 10 by going from
% 4.6037 to      27.6788.
```